Kernix 0.12.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

46 workloads

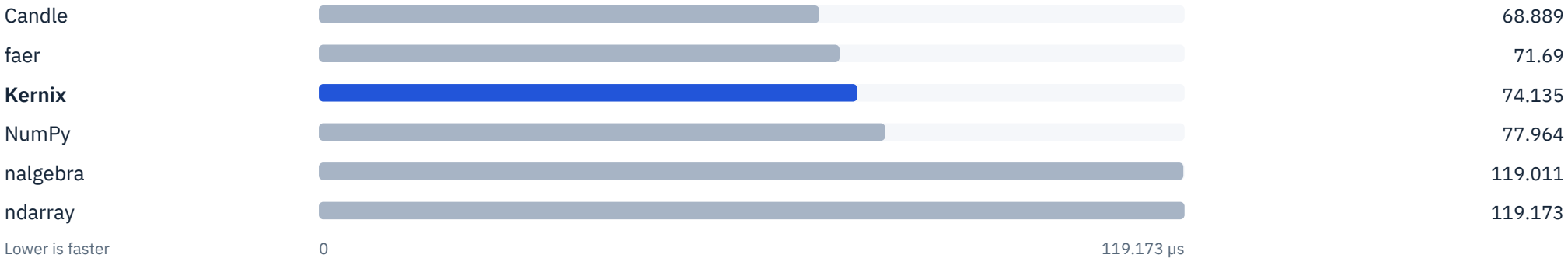
152 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



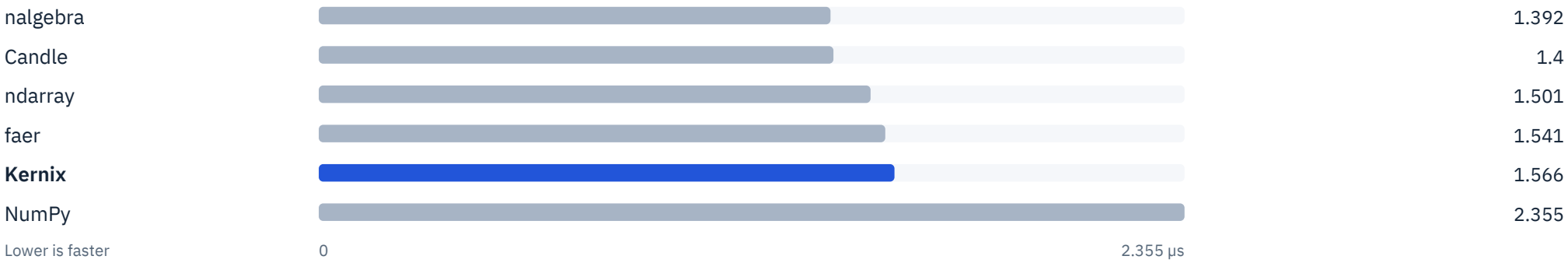
Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

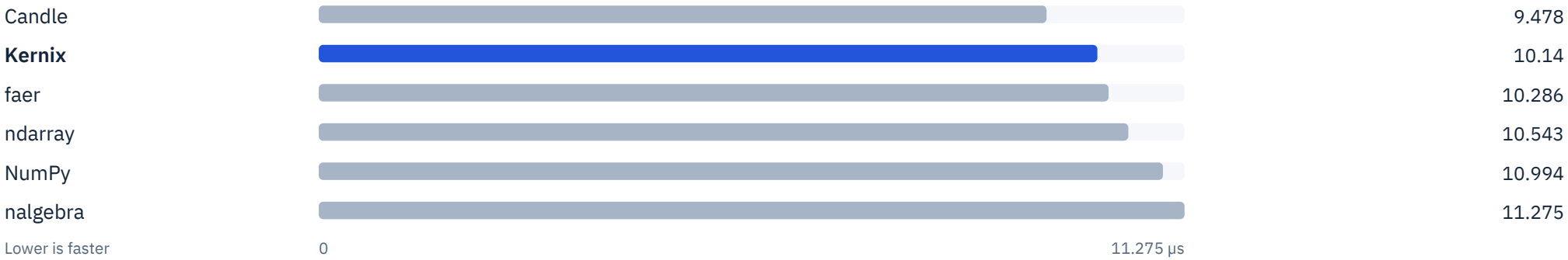
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

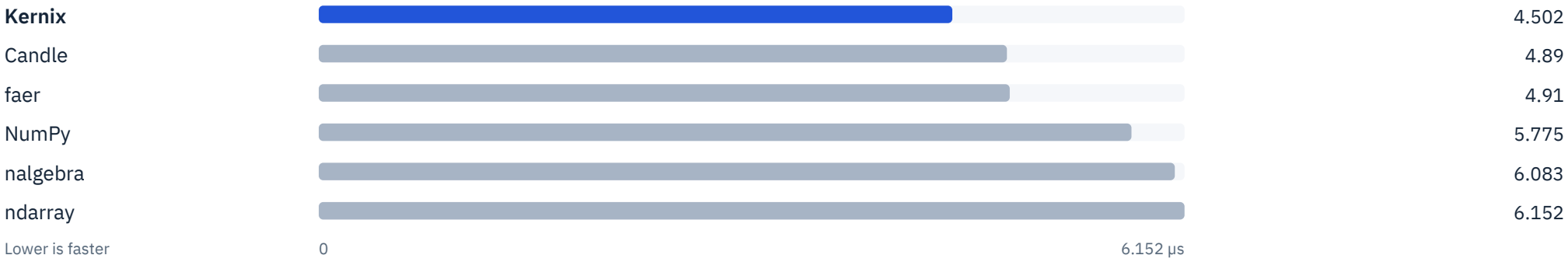
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



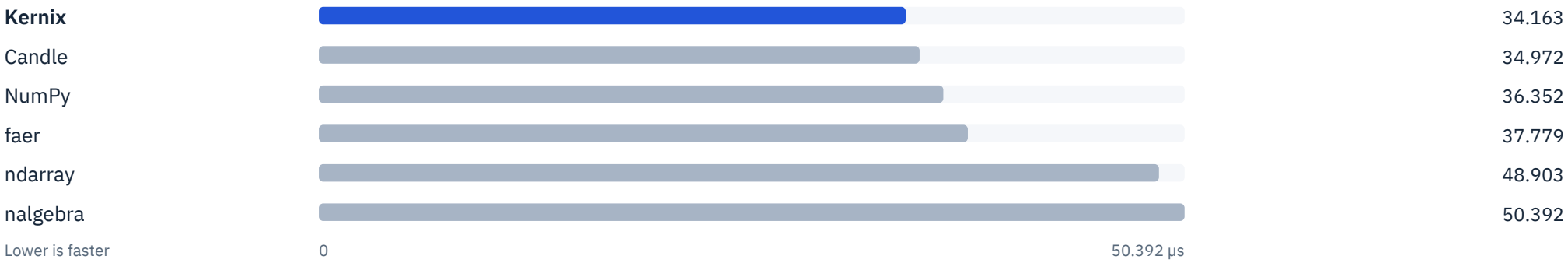
Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

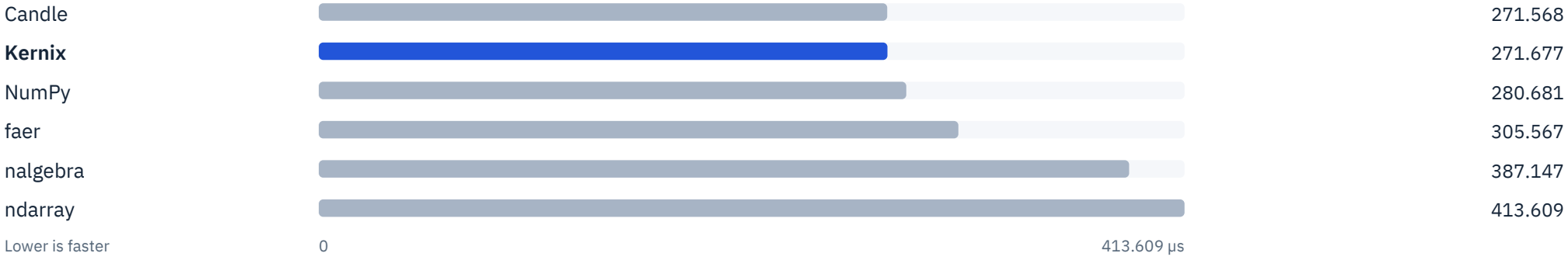
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

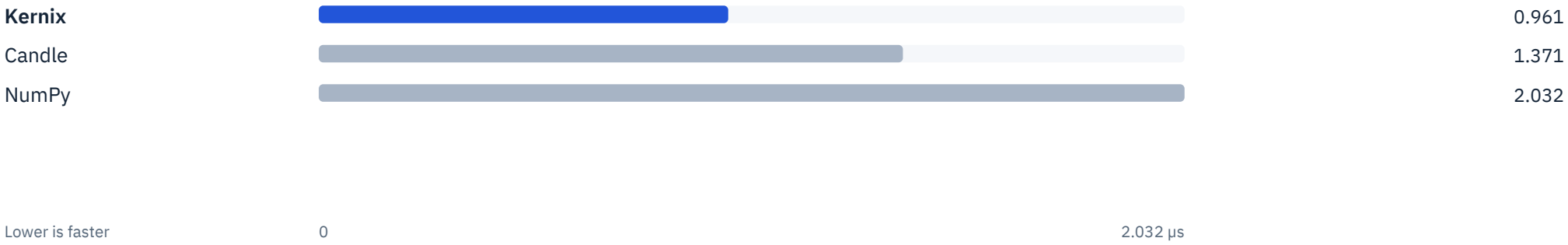
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



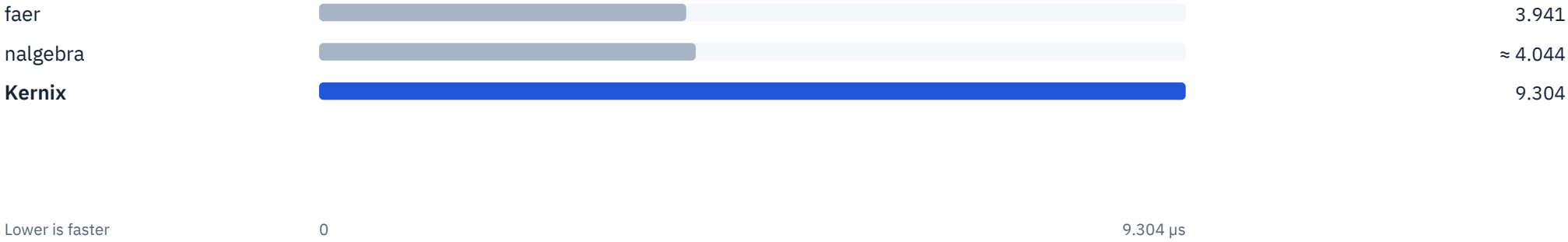
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



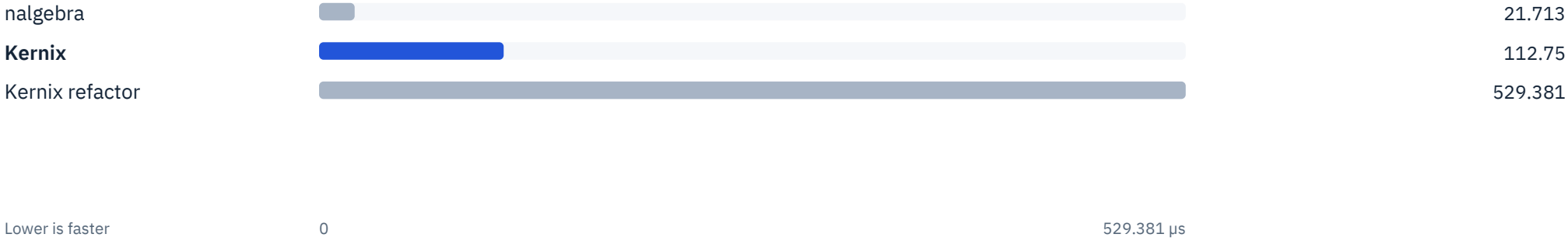
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

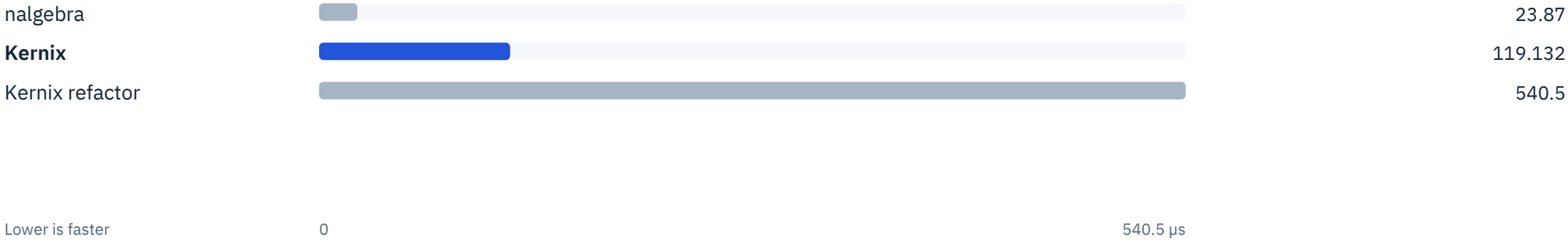
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



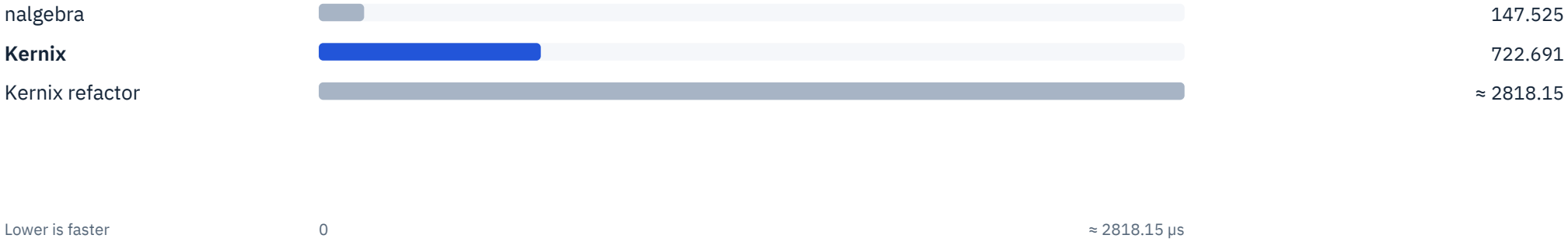
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

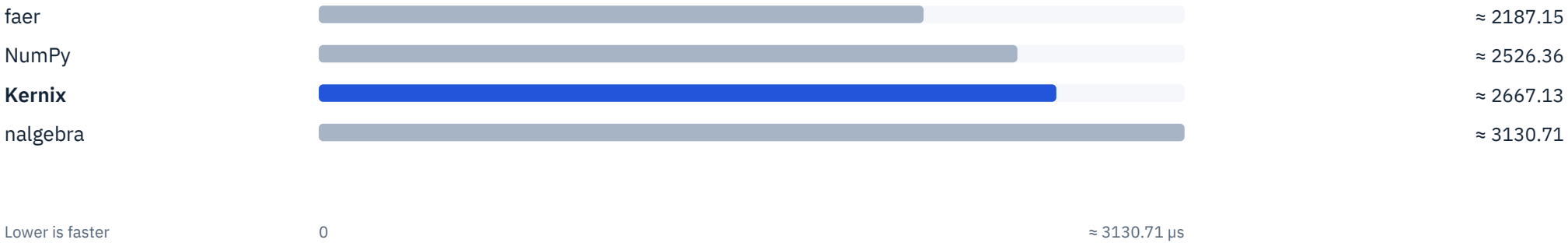
512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

Per-call median

512 × 128 · f64 · 1 worker · allocated output



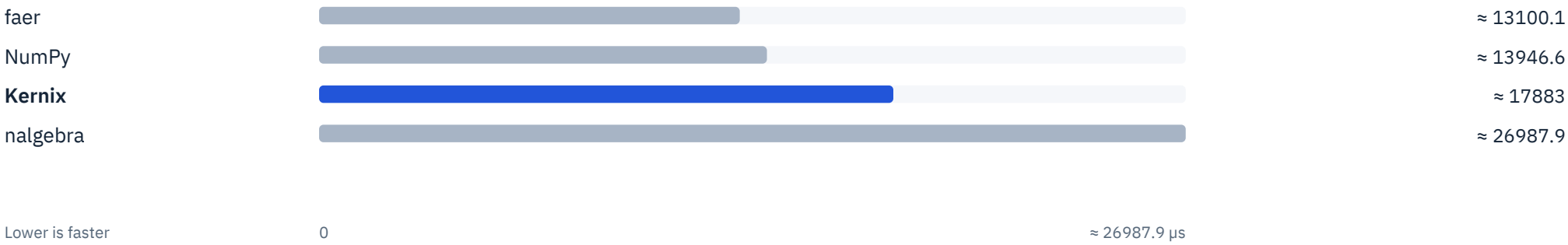
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

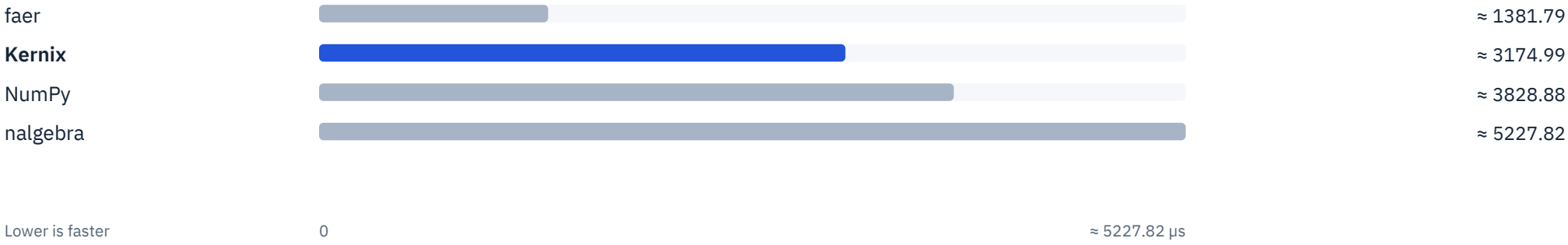
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



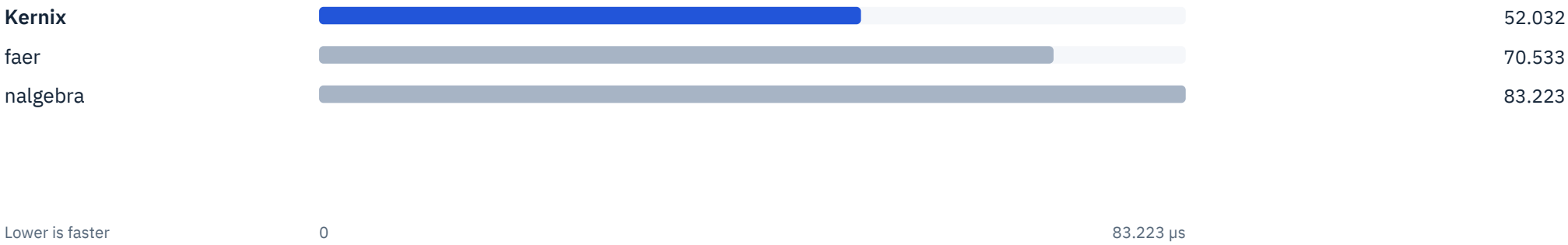
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

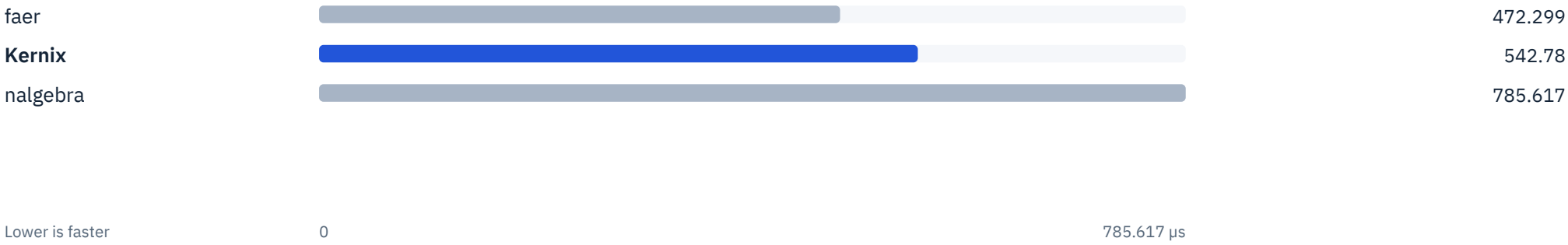
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



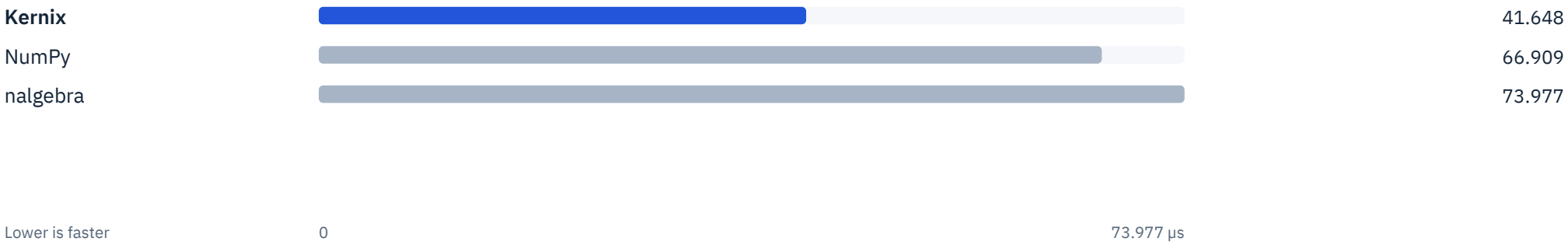
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

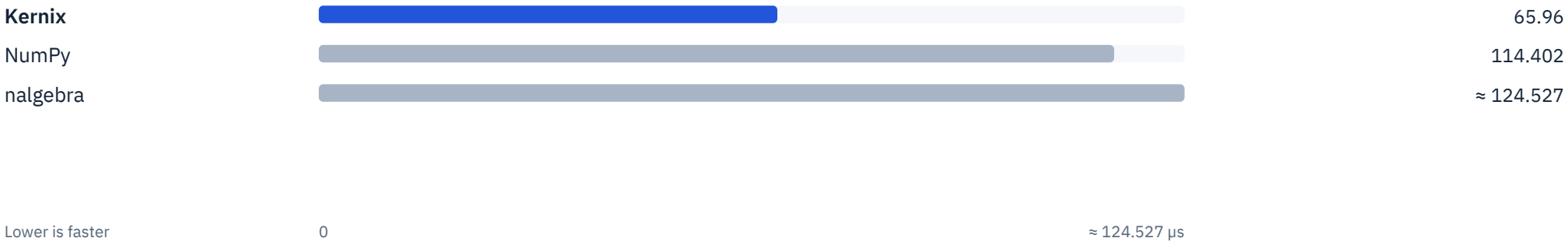
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



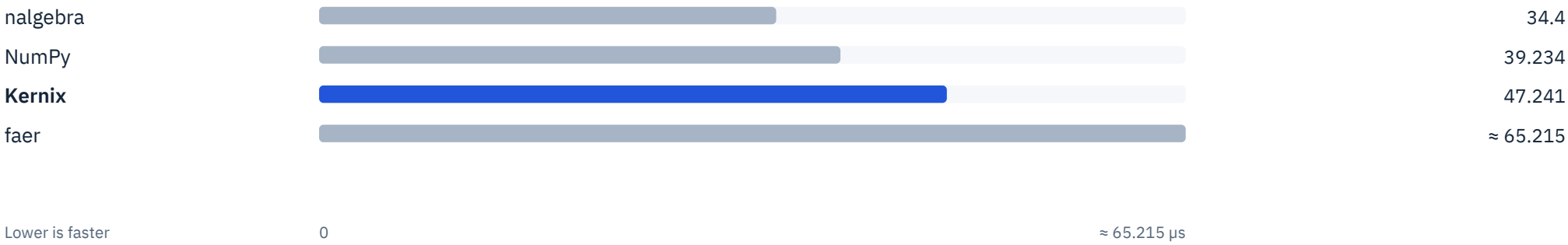
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output

Per-call median

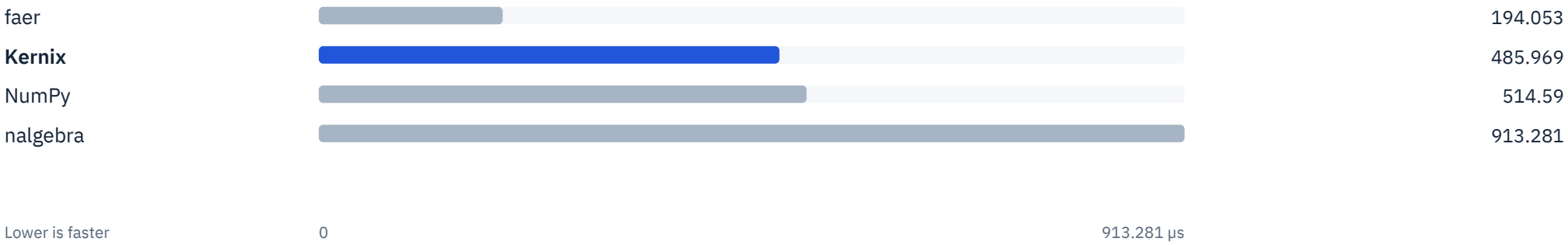


Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



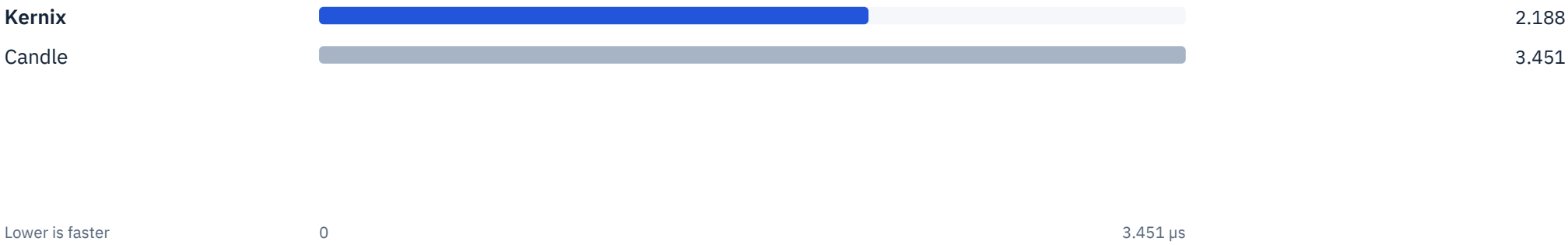
Kernix 0.12.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



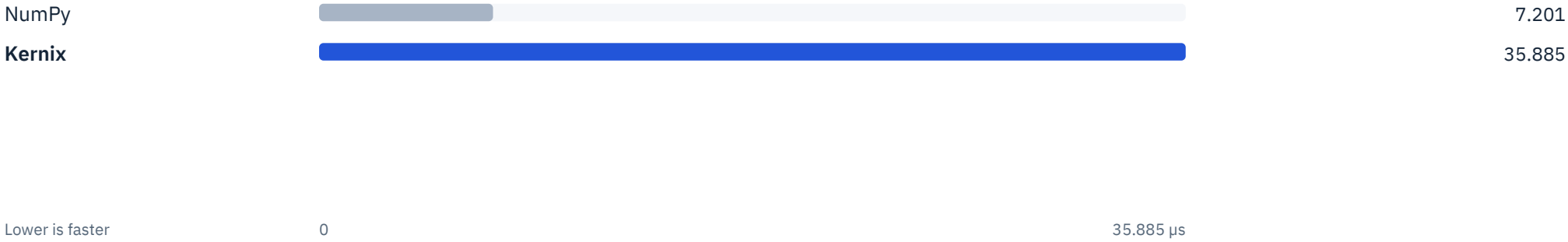
Kernix 0.12.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

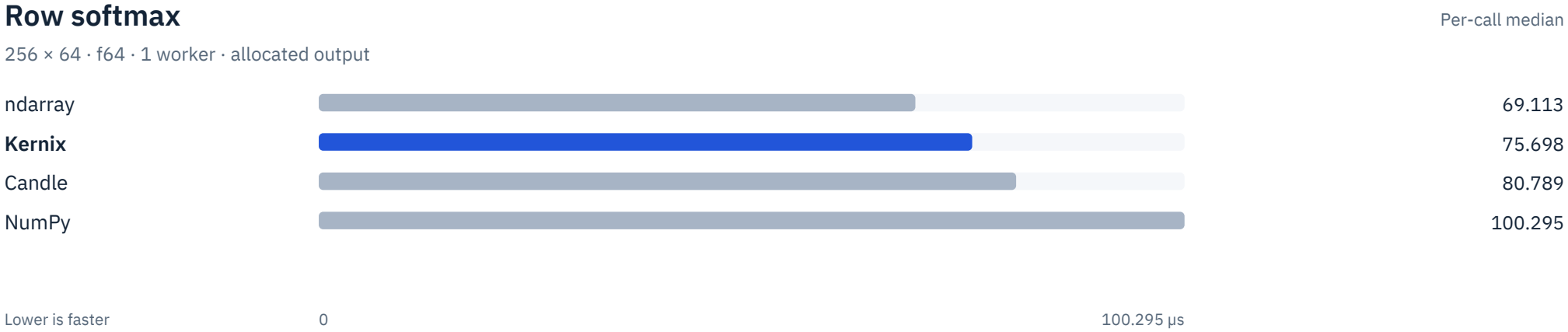


Kernix 0.12.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



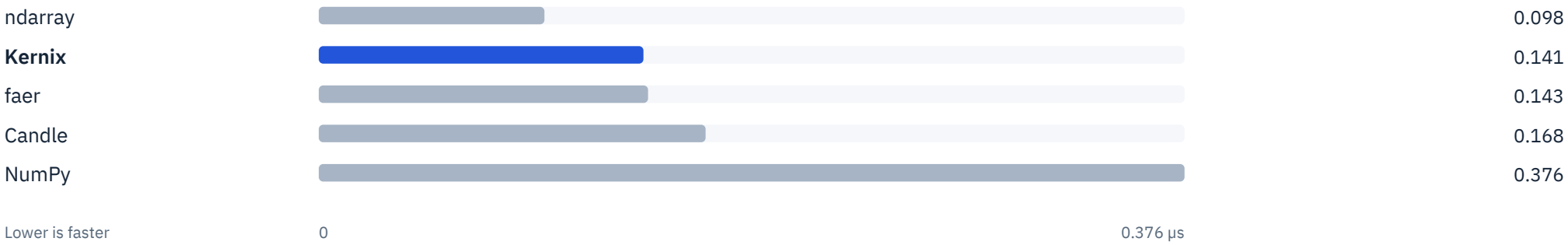
Kernix 0.12.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8549598323436954	0.8378494623655914 – 0.8691263176701804

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.3604773282010496	0.35471006182411524 – 0.38053385725526634

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.44287614487737254	0.4398298429214799 – 0.4580180849626527

Thin QR

512 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7576909363946751	0.7432072422955074 – 0.8038720198704554

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.4991791918264273	0.4803599515307621 – 0.5078611944426149

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9741677808632452	0.9438496742229416 – 0.9984214886844286

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9990550437042287	0.9766783174641437 – 1.0745344105206183

Thin QR

256 × 64 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9788347720307858	0.9634141448186757 – 0.989606890943739

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0133164235890932	0.9890243902439024 – 1.1006211180124224

Matrix multiplication

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9941706698332163	0.9845926337875073 – 0.9993919658419922

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0183357601251146	0.993922368385511 – 1.0356599674090168

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0054575241831976	0.981300896638036 – 1.025779538880808

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9973555103077842	0.9729474862969293 – 0.9986107093239619

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9895963379109447	0.9548841059602649 – 1.0287800687285222

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0028067202414839	0.9733149389042176 – 1.0364563187268607

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0011050291676304	0.9775446246082573 – 1.0589875111507583

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.1025043680838673	0.9660265878877399 – 1.1978680879413723

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0519618239660657	1.006554307116105 – 1.1666666666666667

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.995463411620374	0.9778598743749026 – 1.0276817508382503

Cholesky factorization

256 × 256 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.01549232141547	1.0091252328428177 – 1.031882568924252

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

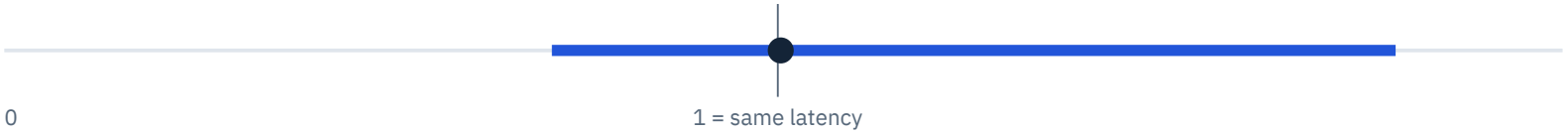


CURRENT / BASELINE	95% INTERVAL
1.00042872342826	0.9864813132222578 – 1.0167709594881083

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.003797530380243	0.7079180180333158 – 1.7986926866237212

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

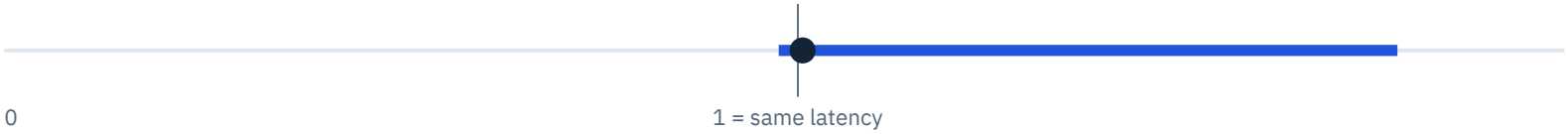


CURRENT / BASELINE	95% INTERVAL
0.965063468796831	0.9576436699021517 – 1.0025281357038003

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0060232840382972	0.9759132167367722 – 1.7554905490549055

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	122731	122.731
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	78575	78.575
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	49265	49.265
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	36336	36.336
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	69063	69.063
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	82268	82.268
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	36784	36.784
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	39052	39.052
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1400	1.4
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2355	2.355

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1501	1.501
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1541	1.541
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1392	1.392
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1566	1.566
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10286	10.286
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11275	11.275
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9478	9.478
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10543	10.543
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	10994	10.994
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	10140	10.14

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	74135	74.135
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	119011	119.011
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	71690	71.69
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	68889	68.889
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	119173	119.173
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	77964	77.964
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	679925	679.925
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	631138	631.138
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	557881	557.881
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	666647	666.647

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	563193	563.193
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	585325	585.325
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4502	4.502
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4890	4.89
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	5775	5.775
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	4910	4.91
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6083	6.083
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6152	6.152
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	34163	34.163
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	48903	48.903

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	37779	37.779
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	50392	50.392
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	36352	36.352
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	34972	34.972
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	387147	387.147
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	271677	271.677
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	305567	305.567
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	280681	280.681
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	413609	413.609
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	271568	271.568

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2756	2.756
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2004.9999999999998	2.0049999999999998
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2222	2.222
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	961	0.961
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2032	2.032
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1371	1.371
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	9656	9.656
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	41991	41.991
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	306270	306.27
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	22457	22.457

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	3941	3.941
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	4043.9999999999995	4.0439999999999995
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	9304	9.304
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1007406	1007.406
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	769663	769.663
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	199222	199.222
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	9067001	9067.001
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	1658008	1658.008
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	6934999	6934.999
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	754701	754.701

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1103033	1103.033
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	239854	239.854
Cholesky rank-one update / nalgebra 256 × 256 · f64	21713	21.713
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	529381	529.381
Cholesky rank-one update / Kernix 256 × 256 · f64	112750	112.75
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	23870	23.87
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	540500	540.5
Cholesky rank-one downdate / Kernix 256 × 256 · f64	119132	119.132
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	2818152	2818.152
Cholesky rank-one update / nalgebra 512 × 512 · f64	147525	147.525

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix 512 × 512 · f64	722691	722.691
Thin QR / Kernix 512 × 128 · f64	2667134	2667.134
Thin QR / nalgebra 512 × 128 · f64	3130708	3130.708
Thin QR / NumPy 512 × 128 · f64	2526361	2526.361
Thin QR / faer 512 × 128 · f64	2187147	2187.147
Thin QR / faer 1024 × 256 · f64	13100052	13100.052
Thin QR / NumPy 1024 × 256 · f64	13946581	13946.581
Thin QR / nalgebra 1024 × 256 · f64	26987916	26987.916
Thin QR / Kernix 1024 × 256 · f64	17883006	17883.006
Cholesky factorization / NumPy 512 × 512 · f64	3828883	3828.883

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 512 × 512 · f64	5227821	5227.821
Cholesky factorization / faer 512 × 512 · f64	1381789	1381.789
Cholesky factorization / Kernix 512 × 512 · f64	3174989	3174.989
Column-pivoted QR / faer 96 × 32 · f64	70533	70.533
Column-pivoted QR / Kernix 96 × 32 · f64	52032	52.032
Column-pivoted QR / nalgebra 96 × 32 · f64	83223	83.223
Column-pivoted QR / nalgebra 256 × 64 · f64	785617	785.617
Column-pivoted QR / Kernix 256 × 64 · f64	542780	542.78
Column-pivoted QR / faer 256 × 64 · f64	472299	472.299
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	41648	41.648

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	73977	73.977
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	66909	66.909
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	124527.00000000001	124.52700000000001
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	65960	65.96
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	114402	114.402
Thin QR / nalgebra 96 × 32 · f64	34400	34.4
Thin QR / NumPy 96 × 32 · f64	39234	39.234
Thin QR / faer 96 × 32 · f64	65215.00000000001	65.21500000000001
Thin QR / Kernix 96 × 32 · f64	47241	47.241
Thin QR / NumPy 256 × 64 · f64	316185	316.185

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	372457	372.457
Thin QR / faer 256 × 64 · f64	426410	426.41
Thin QR / Kernix 256 × 64 · f64	515956.00000000006	515.95600000000006
Cholesky factorization / faer 64 × 64 · f64	8370	8.37
Cholesky factorization / Kernix 64 × 64 · f64	21617	21.617
Cholesky factorization / nalgebra 64 × 64 · f64	14128	14.128
Cholesky factorization / NumPy 64 × 64 · f64	17409	17.409
Cholesky factorization / faer 128 × 128 · f64	37224	37.224
Cholesky factorization / NumPy 128 × 128 · f64	68714	68.714
Cholesky factorization / Kernix 128 × 128 · f64	112836	112.836

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 128 × 128 · f64	96496	96.496
Cholesky factorization / Kernix 256 × 256 · f64	485969	485.969
Cholesky factorization / nalgebra 256 × 256 · f64	913281	913.281
Cholesky factorization / NumPy 256 × 256 · f64	514590	514.59
Cholesky factorization / faer 256 × 256 · f64	194053	194.053
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3870	3.87
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6062	6.062
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4153	4.153
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2254	2.254
Model inference / Candle batched normalized/sliced MLP · f64	3451	3.451

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Kernix batched normalized/sliced MLP · f64	2188	2.188
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2700	2.7
Model recording + forward / Candle batched normalized/sliced MLP · f64	5060	5.06
Model backward / Kernix prepared graph → fresh gradients · f64	2943	2.943
Model backward / Candle prepared graph → fresh gradients · f64	14463	14.463
Parameter update / Kernix transactional SGD, fixed gradients · f64	510	0.51
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	35885	35.885
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	7201	7.201
Row softmax / Kernix 256 × 64 · f64	75698	75.698
Row softmax / ndarray 256 × 64 · f64	69113	69.113

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	100295	100.295
Row softmax / Candle 256 × 64 · f64	80789	80.789
Create zeros / NumPy 64 × 64 · f32	376	0.376
Create zeros / ndarray 64 × 64 · f32	98	0.098
Create zeros / Kernix 64 × 64 · f32	141	0.141
Create zeros / faer 64 × 64 · f32	143	0.143
Create zeros / Candle 64 × 64 · f32	168	0.168
Create ones / faer 64 × 64 · f32	203	0.203
Create ones / Candle 64 × 64 · f32	218	0.218
Create ones / NumPy 64 × 64 · f32	1331	1.331

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / ndarray 64 × 64 · f32	140	0.14
Create ones / Kernix 64 × 64 · f32	201	0.201